Tax Evasion and Productivity

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Tax evasion through input overreporting

- VAT allows firms to deduct taxes paid on their purchases from the taxes collected on sales
- By overreporting their inputs, firms can reduce their tax bill
- Input overreporting is widespread: a common practice and a concern among OECD countries (OECD 2017)
  - Losses of 1.2–2.7% of total tax revenue in Chile, Colombia, Ecuador, Poland and Slovakia
  - Conditional on detected evasion: true losses are plausibly larger
- Governments have limited resources, and overreporting costs the government tax revenue

Question: Can we detect and measure tax evasion through input overreporting? How much does overreporting of inputs cost the government?

Why is it difficult to detect?

- Overreporting is unobserved: firms conceal their actions because it is illegal
- Tax-evading firms might *pass* as low-productivity firms
- For a given level of output, high input use by a firm might be because the firm is overreporting or because its productivity is low

What have people done?

- **Transaction-level data and administrative records** (Carrillo et al. 2022; Zumaya et al. 2021)
  - False invoicing, ghost-firm invoices, invoice mills
  - Access to this data is restricted; it is conditional on detected evasion
- **On the consumer side:** a benchmark group to measure misreporting from potentially evading individuals (Pissarides and Weber 1989; Gorodnichenko et al. 2009)
  - Self-employed vs. employees
  - The self-employed might underreport their income to pay less income tax; employees cannot misreport because their income is reported by their employers
  - Consumption is a function of true income: comparing their consumption backs out how much the self-employed underreport

The natural analogue for firms: output

- Output is a function of true inputs (assuming output is correctly reported)
- If we knew the production function (PF) and productivity, we could back out true inputs: the difference between reported and true inputs is the overreporting
- However, we rely on *reported inputs* to recover the PF and productivity
  - Proxy-variable methods rely on the *flexible* input (Levinsohn and Petrin 2003; Ackerberg et al. 2015; Gandhi et al. 2020)
    - If the flexible input is deductible, firms might overreport it: materials, in my application
- We have to solve the problem simultaneously, for overreporting and productivity

What do I do?

- Instead of looking into the PF equation, I take the FOC of the firm's profit-maximization problem
- The FOC links the materials share of revenue to the materials elasticity of output, up to measurement error
  - It does not depend on productivity (Hicks-neutral productivity)
- Just like in the individuals' case, I estimate this elasticity from a benchmark group of firms that do not overreport
  - The benchmark group can have different productivity than the potentially tax-evading firms
  - The critical assumption: both share a common technology
- With the elasticity at hand, the difference between reported and true materials reveals the overreporting, up to measurement error

What do I find?

Colombian manufacturing, 1981–1991:

1. **Detect:** overreporting in 9 of the 18 industries liable for the sales tax (about two fifths of manufacturing output); none in the 2 exempt food industries
2. **Measure:** potentially tax-evading firms overreport on average 4.5–23.1% of their true materials, depending on the industry
3. **Cost:** undetected overreporting costs 4.7% of gross sales-tax revenue; a 1% increase in the rate on purchases lowers net sales-tax revenue by about 1.7% of gross sales-tax revenue
4. **Productivity:** failing to correct for overreporting leads to biased estimates of the PF and productivity

Appendix

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